Lab 6

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Part B. Computer analysis in R

1. Load and inspect the data.

```
library(gapminder)

d <- subset(gapminder, year == 2007)
d$log_gdp <- log(d$gdpPercap)
d$x <- d$log_gdp - mean(d$log_gdp)

summary(d[, c("lifeExp", "gdpPercap", "log_gdp")])
```

```
    lifeExp        gdpPercap          log_gdp
 Min.   :39.61   Min.   :  277.6   Min.   : 5.626
 1st Qu.:57.16   1st Qu.: 1624.8   1st Qu.: 7.393
 Median :71.94   Median : 6124.4   Median : 8.720
 Mean   :67.01   Mean   :11680.1   Mean   : 8.616
 3rd Qu.:76.41   3rd Qu.:18008.8   3rd Qu.: 9.799
 Max.   :82.60   Max.   :49357.2   Max.   :10.807
```

```
plot(d$log_gdp, d$lifeExp, pch = 19, col = "#2B78A6",
     xlab = "log(GDP per capita)", ylab = "Life expectancy (years)")
```

2. Fit the conjugate model. The next code block implements the normal–inverse-gamma posterior directly. We use weak but proper priors:

$$m_0 = (65, 0)^\top, \qquad V_0 = \mathrm{diag}(100, 100), \qquad a_0 = 2, \qquad b_0 = 100.$$

```r
y <- d$lifeExp
X <- cbind(1, d$x)
n <- nrow(X)

m0 <- c(65, 0)
V0 <- diag(c(100, 100))
a0 <- 2
b0 <- 100

Vn <- solve(solve(V0) + crossprod(X))
mn <- Vn %*% (solve(V0) %*% m0 + crossprod(X, y))
an <- a0 + n / 2
bn <- as.numeric(b0 + 0.5 * (
  crossprod(y) + t(m0) %*% solve(V0) %*% m0 -
  t(mn) %*% solve(Vn) %*% mn
))

mn
```

```
          [,1]
[1,] 67.007281
[2,]  7.202524
```

```r
c(an = an, bn = bn)
```

```
      an       bn
  73.000 3651.136
```

3. Draw from the joint posterior. Here, `1/rgamma(...)` gives inverse-gamma draws under the convention used in class.

```r
set.seed(402)
S <- 10000
sigma2_draw <- 1 / rgamma(S, shape = an, rate = bn)

R <- chol(Vn)
beta_draw <- matrix(mn, nrow = S, ncol = 2, byrow = TRUE) +
  (matrix(rnorm(2 * S), nrow = S, ncol = 2) %*% R) *
  sqrt(sigma2_draw)

colnames(beta_draw) <- c("intercept", "slope")
apply(beta_draw, 2, quantile, probs = c(.025, .5, .975))
```

```
      intercept    slope
2.5%   65.81044 6.320846
50%    66.99885 7.208212
97.5%  68.19318 8.070102
```

```r
mean(beta_draw[, "slope"] > 0)
```

```
[1] 1
```

4. Posterior and predictive summaries at $20,000 GDP per capita.

```r
g_star <- 20000
x_star <- log(g_star) - mean(d$log_gdp)
x_star_vec <- c(1, x_star)

mu_star_draw <- as.vector(beta_draw %*% x_star_vec)
y_star_draw <- rnorm(S, mean = mu_star_draw, sd = sqrt(sigma2_draw))

quantile(mu_star_draw, c(.025, .5, .975))
```

```
    2.5%      50%    97.5%
74.68113 76.28301 77.87913
```

```r
quantile(y_star_draw, c(.025, .5, .975))
```

```
    2.5%      50%    97.5%
61.93719 76.21110 90.94310
```

Part C. What should you report?

1. The 95% credible interval for the slope is (6.32, 8.07). This says that there is a 95% probability that for every 1 unit increase in log GDP per capita, the expected life expectancy increases by 6.32 - 8.07 years.
2. Estimate $Pr(\beta_1 > 0 \mid y)$ from the posterior draws. What does this posterior probability summarize?

```r
mean(beta_draw[, "slope"] > 0)
```

```
[1] 1
```

This posterior probability summarizes how certain we are that the slope is positive, which, in our case, exceeds 0.9999.

3. The 95% credible interval for the conditional mean life expectancy at GDP per capita equal to $20,000 is the 2.5th and 97.5th quantiles in the part 4 code block from above.

For the conditional mean, it is (74.7, 77.9); for a new country, it is (61.9, 90.9). They are different because the credible interval for the mean is the average life expectancy of all countries with the uncertainty in β, but the predictive interval is for one specific country, which includes the uncertainty in β and the uncertainty for an individual country.

4. Make a plot of the data with the posterior mean regression line:

```
plot(d$log_gdp, y, pch = 19, col = "#2B78A6",
     xlab = "log(GDP per capita)", ylab = "Life expectancy (years)")
xx <- seq(min(d$log_gdp), max(d$log_gdp), length.out = 150)
lines(xx, mn[1] + mn[2] * (xx - mean(d$log_gdp)),
      lwd = 3, col = "#143D59")
```

This posterior mean line captures an overall upward positive trend.

Part D. Diagnostics and model limitations

1. Make a residual-versus-fitted plot and a normal Q–Q plot:

```
fitted_mean <- as.vector(X %*% mn)
resid <- y - fitted_mean

plot(fitted_mean, resid, pch = 19, col = "#2B78A6",
     xlab = "Fitted mean life expectancy", ylab = "Residual")
abline(h = 0, lty = 2)
```

```
qqnorm(resid, pch = 19, col = "#2B78A6")
qqline(resid, col = "#143D59", lwd = 2)
```

2. Identify one or two features of the data that make a single Gaussian line imperfect. Consider curvature, unequal residual spread, and countries far from the main trend.

Feature 1: There is a cluster of countries that sit below the line at log GDP, which is also shown in the QQ plot where the lower tail bends far below the line, so the residuals are left-skewed, not normal.

Feature 2: The residual spread is also not constant. The residuals for countries with fitted values above 75 are within +- 5 years. Therefore, the variance overstates the uncertainty for wealthy countries.

3. Fit the same model using *raw* GDP per capita instead of log GDP per capita. Compare its scatterplot and residual plot with the log-GDP version. Which predictor scale seems more appropriate here, and why?

```
d$g <- d$gdpPercap - mean(d$gdpPercap)
X2 <- cbind(1, d$g)

Vn2 <- solve(solve(V0) + crossprod(X2))
mn2 <- Vn2 %*% (solve(V0) %*% m0 + crossprod(X2, y))
mn2
```

```
              [,1]
[1,] 6.700728e+01
[2,] 6.371341e-04
```

```
plot(d$gdpPercap, y, pch = 19, col = "#2B78A6",
     xlab = "GDP per capita", ylab = "Life expectancy (years)")
gg <- seq(min(d$gdpPercap), max(d$gdpPercap), length.out = 150)
lines(gg, mn2[1] + mn2[2] * (gg - mean(d$gdpPercap)), lwd = 3, col = "#143D59")
```

```
resid2 <- y - as.vector(X2 %*% mn2)
plot(as.vector(X2 %*% mn2), resid2, pch = 19, col = "#2B78A6",
     xlab = "Fitted mean life expectancy", ylab = "Residual")
abline(h = 0, lty = 2)
```

The slope is 6.37 x 10^(-4). The scatterplot is strongly curved, with most countries bunched on the left. The points do not match the linear line. The raw residual plot also has a strong arch, with a minor inverted U-shape. Therefore, the log-GDP model would be more appropriate because it spreads the countries more evenly, gives a roughly linear relationship, and the residuals show no obvious pattern.

4. Add a quadratic term in centered log GDP or include continent indicators. What scientific pattern might this richer model capture that the simple line misses?

```r
X3 <- cbind(1, d$x, d$x^2)
m0_3 <- c(65, 0, 0)
V0_3 <- diag(100, 3)

Vn3 <- solve(solve(V0_3) + crossprod(X3))
mn3 <- Vn3 %*% (solve(V0_3) %*% m0_3 + crossprod(X3, y))
mn3
```

```
          [,1]
[1,] 67.874654
[2,]  7.104677
[3,] -0.475117
```

```r
plot(d$log_gdp, y, pch = 19, col = "#2B78A6",
     xlab = "log(GDP per capita)", ylab = "Life expectancy (years)")
lines(xx, mn3[1] + mn3[2] * (xx - mean(d$log_gdp)) + mn3[3] * (xx - mean(d$log_gdp))^2,
      lwd = 3, col = "#143D59")
```

Adding a quadratic term gives posterior means $\beta_0 = 67.88$, $\beta_1 = 7.11$, $\beta_2 = -0.48$. The negative quadratic coefficient means the relationship is concave. This captures diminishing returns to income that the simple line misses. For poor countries, extra income buys large gains such as basic healthcare, sanitation, and nutrition, while wealthy countries are already near a biological ceiling of around 80 years, so further income adds less. The simple line uses a single slope of 7.2 everywhere, which underpredicts middle-income countries and overpredicts the richest ones.